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On Contractible Open Topological Manifolds

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1. Introduction

We show that a contractible, at infinity 1-connected, open n-dimensional topological manifold is homeomorphic to the n-dimensional euclidean space $\mathbb{R}^n$ for $n \geqq 5$.

The analogous theorem in the category of piecewise linear manifolds and hence in the category of differentiable manifolds (by virtue of [6]) was proved by STALLINGS in [9]: If $n \geqq 5$ every contractible, at infinity 1-connected, open n-dimensional piecewise linear (differentiable) manifold is piecewise linearly homeomorphic (diffeomorphic) to the n-dimensional euclidean space $\mathbb{R}^n$ with its standard piecewise linear (differentiable) structure. The proof in [9] is based on an engulfing theorem for piecewise linear manifolds.

Our proof of the corresponding version for topological manifolds is similar in spirit to the proof of the piecewise linear case in [9]. We apply an engulfing theorem for topological manifolds proved recently in [7].

This topological engulfing theorem was applied in [7] to prove the topological Poincaré conjecture in dimensions $n \geqq 5$: If $n \geqq 5$ every n-dimensional topological manifold which is a homotopy n-sphere is a topological n-sphere.

We add the following result in dimension 4: If a 4-dimensional topological manifold is a homotopy 4-sphere, then its suspension is a topological 5-sphere. (Compare [2] for the differentiable case.)

We would like to thank the referee for pointing out some flaws in the original version of this paper.

2. Preliminaries

An n-dimensional topological manifold M^n is a separable Hausdorff space such that each point has an open neighborhood homeomorphic to an open subset of $\mathbb{R}^n$ or to an open subset of a closed half-space of $\mathbb{R}^n$. The points of M^n which do not have an open neighborhood homeomorphic to an open subset of $\mathbb{R}^n$ form the boundary ∂M^n of M^n. If the boundary is empty, M^n is said to be a topological manifold without

boundary. A topological manifold without boundary is called closed, if it is compact, and it is called open, if it is not compact.

A topological space T is said to be k-connected, if the homotopy groups $\pi_i(T)=0$ for $0\leqq i\leqq k$. It is said to be 1-connected at infinity, if for each compact subset C of T, there is a compact subset D of T with $C\subset D\subset T$, such that $T-D$ is 1-connected.

The image $f(\mathbb{R}^n)$ of $\mathbb{R}^n$, where $f\colon \mathbb{R}^n\to T$ is a homeomorphism onto an open subset of the topological space T, is called an open (topological) n-cell in T.

The property of a topological manifold to be contractible can be expressed in terms of its homotopy groups.

Lemma 2.1. *An n-dimensional topological manifold is contractible if and only if it is n-connected.*

Proof. If an n-dimensional topological manifold M^n is n-connected, then $\pi_i(M^n)=0$ for all $i\geqq 0$ by the Hurewicz isomorphism theorem. A topological manifold has the homotopy type of a CW-complex as shown in [5]. But a CW-complex with trivial homotopy groups is contractible. Hence M^n is contractible.

Lemma 2.2 (cf. [9]). *If $M^n=M_1\times M_2$, where M_1 and M_2 are topological manifolds, neither of which is a point, and if M^n is a contractible open n-dimensional topological manifold and $n\geqq 3$, then M^n is 1-connected at infinity.*

Let $C_\rho^n=\{x;\, x=(x_1, \ldots, x_n)\in\mathbb{R}^n$ with $|x_i|\leqq\rho$ for $i=1, \ldots, n\}$ be the cube in $\mathbb{R}^n$ with side length 2ρ and with the origin as centre.

The following notation was introduced in [7]: Let M^n be an n-dimensional topological manifold. A subset $X\subset M^n$ is called k-dominated, if there are homeomorphisms $g_i\colon C_1^n\to M^n$ into M^n with $\{g_i(\mathrm{int}(C_1^n))\}$ covers X and with $g_i^{-1}(g_i(C_1^n)\cap X)$ is contained in a k-dimensional subpolyhedron of C_1^n for each index i.

Engulfing Theorem (cf. NEWMAN [7]). *Let M^n be an n-dimensional topological manifold without boundary, let $X\subset M^n$ be closed, and let $V\subset M^n$ be open. Suppose that M^n is k-connected, V is $(k-1)$-connected, X is k-dominated, and $X-V$ is compact. Assume that $k\leqq n-3$.*

Then there is a homeomorphism $h\colon M^n\to M^n$ onto M^n with $X\subset h(V)$ and with $h|M^n-C=\mathrm{id}|M^n-C$ for a compact subset $C\subset M^n$.

(In [7] actually a stronger result was proved: The homeomorphism h, in addition, is isotopic to the identity, the homeomorphisms of the isotopy being the identity outside the compact subset C. For our purposes, however, this additional property of the homeomorphism h will not be required.)

3. The Main Results

The following lemma was essentially proved in [7], where it provided the main argument in the proof of the topological Poincaré conjecture in dimensions $n \geqq 5$. For our purposes, however, we need some slight modifications.

$|K|$ denotes the set of points of a simplicial complex K.

Lemma 3.1 (cf. [7], Theorem 7). *Let M^n be a contractible open n-dimensional topological manifold, and let $g_i\colon C_1^n \to M^n$, $i=1, 2$, and $g\colon C_1^n \to M^n$ be homeomorphisms into M^n. We consider a fixed metric d on M^n. Suppose that $\varepsilon>0$, α, β, and γ are given, where $0<\alpha<1$, and $0<\beta<\gamma<1$.*

If $n \geqq 5$ there are homeomorphisms $h_i\colon M^n \to M^n$ onto M^n, $i=1, 2$, such that:

(1) $g_1(C_\beta^n) \cup g_2(C_\beta^n) \cup g(C_\alpha^n) \subset h_1\big(g_1(C_\gamma^n)\big) \cup h_2\big(g_2(C_\gamma^n)\big)$.

(2) $h_1 | g_1(C_\beta^n) = \mathrm{id} | g_1(C_\beta^n)$.

(3) $h_2 | g_2(C_\beta^n) - g(C_1^n) = \mathrm{id} | g_2(C_1^n) - g(C_\beta^n)$ *and* $d\big(h_2(p), p\big) < \varepsilon$ *for* $p \in g_2(C_\beta^n)$.

Proof. We choose α' and β' with $0<\alpha<\alpha'<1$ and with $0<\beta<\beta'<\gamma<1$. Let K be the simplicial complex determined by a simplicial subdivision of the cube $C_{\alpha'}^n$ such that:

(a) C_α^n forms the set of points of a subcomplex of K.

(b) The diameter of $g(\Delta)$ is less than ε for each simplex Δ of K.

(c) If $g(\Delta) \cap g_2(C_\beta^n)$ is not empty for a simplex Δ of K, then $g(\Delta) \subset g_2(C_{\beta'}^n)$.

Let K^{n-3} denote the $(n-3)$-skeleton of K, and let L^2 be the complementary complex of the first barycentric subdivision of K^{n-3} in the first barycentric subdivision of K. (The complementary complex of a subcomplex of a simplicial complex is the set of all those simplexes of the simplicial complex which do not have a face in the subcomplex.) K^{n-3} is $(n-3)$-dimensional and L^2 is 2-dimensional. Since $n \geqq 5$, we have that $2 \leqq n-3$.

We consider the n-dimensional topological manifold $M_1^n = M^n - g_1(C_\beta^n)$. M_1^n is $(n-2)$-connected. (Namely, it is simply connected by VAN KAMPEN'S theorem for example, the singular homology groups are trivial in dimensions $i=1, \ldots, n-2$, and therefore $\pi_i(M_1^n)=0$ for $i=0, 1, \ldots, n-2$ by the Hurewicz isomorphism theorem). The open subset $V_1 = g_1(\mathrm{int}(C_\gamma^n)) - g_1(C_\beta^n)$ of M_1^n is $(n-2)$-connected. Let $X_1 = g(|K^{n-3}|) - g_1(C_\beta^n)$. Then X_1 is closed in M_1^n, $(n-3)$-dominated, and $X_1 - V_1$ is compact. The Engulfing Theorem can be applied. There is a homeomorphism

$h_1'\colon M_1^n \to M_1^n$ onto M_1^n with $X_1 \subset h_1'(V_1)$ and with $h_1'|M_1^n - C_1 = \mathrm{id}|M_1^n - C_1$ for a compact subset $C_1 \subset M_1^n$. We extend the homeomorphism h_1' to a homeomorphism $h_1'\colon M^n \to M^n$ onto M^n by defining $h_1'|g_1(C_\beta^n) = \mathrm{id}|g_1(C_\beta^n)$. Thus we obtain that

$$g(|K^{n-3}|) \subset h_1'(g_1(\mathrm{int}(C_\gamma^n))) \quad \text{and} \quad h_1'|g_1(C_\beta^n) = \mathrm{id}|g_1(C_\beta^n).$$

Next we apply the Engulfing Theorem to the n-dimensional topological manifold $M_2^n = M^n - g_2(C_{\beta'}^n)$ with respect to the open subset $V_2 = g_2(\mathrm{int}(C_\gamma^n)) - g_2(C_{\beta'}^n)$ and the closed subset $X_2 = g(|L^2|) - g_2(C_{\beta'}^n)$. Similarly, there is a homeomorphism $h_2'\colon M^n \to M^n$ onto M^n such that

$$g(|L^2|) \subset h_2'(g_2(\mathrm{int}(C_\gamma^n))) \quad \text{and} \quad h_2'|g_2(C_{\beta'}^n) = \mathrm{id}|g_2(C_{\beta'}^n).$$

We now consider the open subsets $U_1 = g^{-1}(h_1 \cdot g_1(\mathrm{int}(C_\gamma^n))) \cap C_{\alpha'}^n$ and $U_2 = g^{-1}(h_2 \cdot g_2(\mathrm{int}(C_\gamma^n))) \cap C_{\alpha'}^n$ in $C_{\alpha'}^n$. Then $|K^{n-3}| \subset U_1$ and $|L^2| \subset U_2$. Every point $x \in C_{\alpha'}^n - (|K^{n-3}| \cup |L^2|)$ is contained in a unique segment $[u, v]$ lying in an n-simplex of K, where $u \in |K^{n-3}|$ and $v \in |L^2|$. There is a stretching homeomorphism $s\colon C_{\alpha'}^n \to C_{\alpha'}^n$ onto $C_{\alpha'}^n$ defined by mapping each segment piecewise linearly onto itself such that

$$C_\alpha^n \subset U_1 \cup s(U_2) \quad \text{and} \quad s|\partial C_{\alpha'}^n = \mathrm{id}|\partial C_{\alpha'}^n.$$

We lift the homeomorphism s onto the manifold M^n defining $\tilde{s}\colon M^n \to M^n$ by $\tilde{s}(p) = g \cdot s \cdot g^{-1}(p)$ for $p \in g(C_\alpha^n)$ and $\tilde{s}(p) = p$ for $p \in M^n - g(C_{\alpha'}^n)$. Consequently,

$$g(C_\alpha^n) \subset h_1'(g_1(C_\gamma^n)) \cup \tilde{s} \cdot h_2'(g_2(C_\gamma^n)).$$

On the other hand, s moves $x \in C_{\alpha'}^n$ only within a simplex of K. We conclude from property (c) of the simplicial complex K that

$$g_2(C_\beta^n) \subset \tilde{s} \cdot h_2'(g_2(C_\gamma^n)).$$

Let $h_1 = h_1'$ and $h_2 = \tilde{s} \cdot h_2'$. The properties (1), (2), and (3) are satisfied.

Theorem 3.1. *Let M^n be a contractible open n-dimensional topological manifold. If $n \geqq 5$, then M^n can be covered with two open n-cells.*

Proof. Let d be a fixed metric on M^n. We can find homeomorphisms $g_j\colon C_1^n \to M^n$, $j = 1, 2, \ldots$, with $\{g_j(C_1^n)\}$ is a locally finite covering of M^n and with $\{g_j(C_{\frac{1}{2}}^n)\}$ is a covering of M^n.

We consider further a sequence $U_1, U_2, \ldots$ of open sets in M^n with $\overline{U}_j$ is compact, $\overline{U}_j \subset U_{j+1}$, and with

$$\bigcup_{j=1}^{\infty} U_j = M^n.$$

Let δ_j denote the distance between $\overline{U}_j$ and $M^n - U_{j+1}$, $j = 1, 2, \ldots$.

Define $\beta_j=1-2^{-j}$ and $\gamma_j=1-2^{-(j+1)}$, $j=1,2,\ldots$. We apply Lemma 3.1 and define inductively two sequences $f_{i,0}, f_{i,1}, \ldots$, $i=1,2$, of homeomorphisms $f_{i,j}\colon C_1^n \to M^n$ into M^n, where $f_{i,0}=g_i$, $i=1,2$, and further a monotone increasing sequence $m_0, m_1, \ldots$ of positive integers, where $m_0=1$, such that for each $j\geqq 1$ the following holds:

(1) $\bigcup_{k=1}^{j+2} g_k(C^n_{\alpha_k}) \subset f_{1,j}(C^n_{\gamma_j}) \cup f_{2,j}(C^n_{\gamma_j})$.

(2) $f_{1,j}|C^n_{\beta_j} = f_{1,j-1}|C^n_{\beta_j}$.

(3) $f_{2,j-1}(C^n_{\beta_j}) \subset U_{m_j}$,

$$f_{2,j}|C^n_{\beta_j} - (f_{2,j-1})^{-1}(g_{j+2}(C_1^n)) = f_{2,j-1}|C^n_{\beta_j} - (f_{2,j-1})^{-1}(g_{j+2}(C_1^n)),$$

and

$$d(f_{2,j}(x), f_{2,j-1}(x)) < 2^{-j}\cdot \min\{\delta_1, \ldots, \delta_{m_j}\} \qquad \text{for } x \in C^n_{\beta_j}.$$

Namely suppose that $f_{1,j-1}$ and $f_{2,j-1}$, $j\geqq 1$, are constructed, and that m_{j-1} is given. Since $f_{2,j-1}(C^n_{\beta_j})$ is compact, there is an integer $m_j \geqq m_{j-1}$ with $f_{2,j-1}(C^n_{\beta_j}) \subset U_{m_j}$. In Lemma 3.1, let $g_i=f_{i,j-1}$, $i=1,2$, $g=g_{j+2}$, $\alpha=\frac{1}{2}$, $\beta=\beta_j$, $\gamma=\gamma_j$, and $\varepsilon=2^{-j}\cdot\min\{\delta_1,\ldots,\delta_{m_j}\}$. We obtain homeomorphisms $h_i\colon M^n \to M^n$ onto M^n, $i=1,2$, with the properties described in Lemma 3.1. Define $f_{i,j}=h_i\cdot f_{i,j-1}$, $i=1,2$. The properties (1), (2), and (3) are satisfied.

We now define maps

$$f_i\colon \operatorname{int}(C_1^n) \to M^n \quad \text{by} \quad f_i(x)=\lim_{j\to\infty} f_{i,j}(x), \qquad i=1,2.$$

It follows from property (2) of the construction that $f_1|_{C^n_{\beta_{j_0}}} = f_{1,j}|_{C^n_{\beta_{j_0}}}$ for all $j\geqq j_0-1$. Hence, f_1 is well defined, and it is a homeomorphism onto an open subset of M^n.

For each $j_0\geqq 1$, there is an integer $N(j_0)$ such that $f_2|_{C^n_{\beta_{j_0}}} = f_{2,j}|_{C^n_{\beta_{j_0}}}$ for all $j\geqq N(j_0)$. Namely, because $\{g_j(C_1^n)\}$ is a locally finite covering of M^n, there is an integer $N(j_0)$ with $U_{m_{j_0}+1} \cap g_{j+2}(C_1^n)$ is empty for all $j\geqq N(j_0)$. It follows from property (3) that $f_2|_{C^n_{\beta_{j_0}}} = f_{2,j}|_{C^n_{\beta_{j_0}}}$ for all $j\geqq N(j_0)$. Thus the map $f_2\colon \operatorname{int}(C_1^n) \to M^n$ is well defined and it is a homeomorphism onto an open subset of M^n.

Let $O_i^n = f_i(\operatorname{int}(C_1^n))$, $i=1,2$. Then O_1^n and O_2^n are open n-cells in M^n, and $M^n = O_1^n \cup O_2^n$ by construction.

In the proof of the next theorem, we will use the following notations: Let K be a simplicial complex, and let $S\subset |K|$ be a subset, then $N(S,K)$ denotes the subcomplex of K formed by the simplexes Δ of K with $\Delta \cap S$ is not empty and by the faces of Δ. If K is a simplicial complex, then $\beta^j(K)$ is the jth barycentric subdivision of K.

Theorem 3.2. *Let M^n be a contractible, at infinity 1-connected, open n-dimensional topological manifold. If $n \geqq 5$, then M^n is homeomorphic to the n-dimensional euclidean space $\mathbb{R}^n$.*

Proof. (A) We first show that if $C \subset M^n$ is a compact subset, then there is an open n-cell O^n in M^n with $C \subset O^n$.

By Theorem 3.1, M^n can be covered with two open n-cells O_1^n and O_2^n. Let $f_j\colon \mathbb{R}^n \to M^n$ be homeomorphisms into M^n with $f_j(\mathbb{R}^n) = O_j^n$, $j = 1, 2$.

We consider a fixed simplicial subdivision of $\mathbb{R}^n$ into a simplicial complex K such that:

(a) $C_{i+1}^n - \mathrm{int}(C_i^n)$ is the set of points of a subcomplex of K for each integer $i > 0$.

(b) Let d be a fixed metric on M^n. The diameter of $f_1(\Delta)$ is less than i^{-1} for each simplex Δ of $C_{i+1}^n - \mathrm{int}(C_i^n)$ and for each integer $i > 0$.

(c) If $f_1(\Delta) \cap f_2(C_i^n)$ is not empty for a simplex Δ in C_{i+2}^n, then $f_1(\Delta) \subset f_2(\mathrm{int}(C_{i+1}^n))$.

Since M^n is a normal space, there exists a closed subset $A_1 \subset O_1^n$ with

$$A_1 \cup O_2^n = M^n.$$

We introduce the subcomplexes $L = N(f_1^{-1}(A_1), K)$ of K, $L_1 = \beta^2(N(|\beta^2(L)|, \beta^2(K)))$ of $\beta^4(K)$, and $L_2 = N(|L_1|, \beta^4(K))$ of $\beta^4(K)$. It follows from property (b) of the simplicial complex K that $f_1(|L|)$, $f_1(|L_1|)$, and thus $f_1(|L_2|)$ are closed in M^n.

For each integer $i > 0$ we define

$$K_i = N\big(C_i^n \cap f_1^{-1}(A_1), K\big),$$

$$R_i^n = \beta^2\big(N(|\beta^2(K_i)|, \beta^2(K))\big),$$

and

$$\hat{R}_i^n = N\big(|R_i^n|, \beta^4(K)\big).$$

R_i^n is a regular neighborhood of K_i, and hence a compact piecewise linear manifold with boundary [11]. $\hat{R}_i^n$ is a regular neighborhood of R_i^n, and it is also a piecewise linear manifold with boundary. It is well known that $|\hat{R}_i^n| - \mathrm{int}(|R_i^n|)$ is piecewise linearly homeomorphic to $|\partial R_i^n| \times I$, $I = [0, 1]$ the unit interval. By construction,

$$R_i^n, \hat{R}_i^n \text{ are subcomplexes of } L_2,$$

$$|R_i^n| \subset C_{i+2}^n,$$

$$\mathrm{int}(|R_i^n|) \text{ is an open subset of } \mathbb{R}^n,$$

and

$$A_1 \subset \bigcup_{i=1}^{\infty} f_1\big(\mathrm{int}(|R_i^n|)\big).$$

Let L^2 be the 2-skeleton of L_2. Since M^n is 1-connected at infinity, there is a compact subset $D \subset M^n$ with $C \subset D \subset M^n$ and with $M^n - D$ is 1-connected. We consider the 2-dominated closed subset $X_1 = f_1(|L^2|)$ and the 1-connected open subset $V_1 = M^n - D$ in M^n. Then $X_1 - V_1 = X_1 \cap D_1$ is compact. The Engulfing Theorem can be applied. There is a compact subset $C_1 \subset M^n$, and there is a homeomorphism $h_1 : M^n \to M^n$ onto M^n with

$$f_1(|L^2|) \subset h_1(M^n - D) \quad \text{and} \quad h_1 | M^n - C_1 = \mathrm{id} | M^n - C_1.$$

Since $C_1 \cup D$ is compact, we can find an integer $i > 0$ with

$$C_1 \cup D \subset f_1(|R_i^n|) \cup f_2(C_i^n).$$

Let P^2 denote the 2-skeleton of $\hat{R}_i^n$, and let Q^{n-3} be the complementary complex of the first barycentric subdivision of P^2 in the first barycentric subdivision of $\hat{R}_i^n$. We notice that

$$f_1(|P^2|) \subset h_1(M^n - D).$$

The topological manifold $M_2^n = M^n - f_2(C_{i+1}^n)$ is $(n-2)$-connected. The open subset $V_2 = f_2(\mathrm{int}(C_{i+2}^n)) - f_2(C_{i+1}^n)$ of M_2^n is $(n-2)$-connected, and the subset $X_2 = f_1(|Q^{n-3}|) - f_2(C_{i+1}^n)$ is closed in M_2^n, $(n-3)$-dominated, and $X_2 - V_2$ is compact. We apply once more the Engulfing Theorem. There is a homeomorphism $h_2 : M_2^n \to M_2^n$ onto M_2^n with $X_2 \subset h_2(V_2)$ and with $h_2 | M_2^n - C_2 = \mathrm{id} | M_2^n - C_2$ for a compact subset $C_2 \subset M_2^n$. The homeomorphism h_2 can be extended to a homeomorphism $h_2 : M^n \to M^n$ onto M^n by defining $h_2 | f_2(C_{i+1}^n) = \mathrm{id} | f_2(C_{i+1}^n)$. Hence we obtain that

$$f_1(|Q^{n-3}|) \subset h_2(f_2(\mathrm{int}(C_{i+2}^n))) \quad \text{and} \quad h_2 | f_2(C_{i+1}^n) = \mathrm{id} | f_2(C_{i+1}^n).$$

Next we consider the open subsets $U_1 = f_1^{-1}(h_1(M^n - D)) \cap |\hat{R}_i^n|$ and $U_2 = f_1^{-1}(h_2 \cdot f_2(\mathrm{int}(C_{i+2}^n))) \cap |\hat{R}_i^n|$ in $|\hat{R}_i^n|$. By construction, $|P^2| \subset U_1$ and $|Q^{n-3}| \subset U_2$. In the same way as in Lemma 3.1, a stretching homeomorphism $s : |\hat{R}_i^n| \to |\hat{R}_i^n|$ onto $|\hat{R}_i^n|$ can be found such that s moves a point $x \in |\hat{R}_i^n|$ only within a simplex of $\hat{R}_i^n$ and such that

$$|R_i^n| \subset U_1 \cup s(U_2) \quad \text{and} \quad s | \partial |\hat{R}_i^n| = \mathrm{id} | \partial |\hat{R}_i^n|.$$

We lift the homeomorphism s onto the manifold M^n and define $\tilde{s} : M^n \to M^n$ by $\tilde{s}(p) = f_1 \cdot s \cdot f_1^{-1}(p)$ for $p \in f_1(|\hat{R}_i^n|)$ and $\tilde{s}(p) = p$ for $p \in M^n - f_1(|\hat{R}_i^n|)$. We conclude that

$$f_1(|R_i^n|) \cup f_2(C_i^n) \subset h_1(M^n - D) \cup \tilde{s} \cdot h_2 \cdot f_2(\mathrm{int}(C_{i+2}^n)).$$

(Namely, $f_2(C_i^n) \subset \tilde{s} \cdot h_2 \cdot f_2(\text{int}(C_{i+1}^n))$ because of $|R_i^n| \subset C_{i+2}^n$, property (c) of the simplicial complex K, and the definition of s.) Consequently,

$$C_1 \cup D \subset h_1(M^n - D) \cup \tilde{s} \cdot h_2 \cdot f_2(\text{int}(C_{i+2}^n)).$$

Since $M^n - (C_1 \cup D) = (M^n - D) \cap (M^n - C_1) \subset h_1(M^n - D)$, we finally obtain that

$$M^n = h_1(M^n - D) \cup \tilde{s} \cdot h_2 \cdot f_2(\text{int}(C_{i+2}^n)),$$

or

$$M^n = (M^n - D) \cup h_1^{-1} \cdot \tilde{s} \cdot h_2 \cdot f_2(\text{int}(C_{i+2}^n)).$$

Because $C \subset D$, we must have that

$$C \subset h_1^{-1} \cdot \tilde{s} \cdot h_2 \cdot f_2(\text{int}(C_{i+2}^n)).$$

Let $O^n = h_1^{-1} \cdot \tilde{s} \cdot h_2 \cdot f_2(\text{int}(C_{i+2}^n))$. Then $C \subset O^n$.

(B) It follows from (A) that there is a a sequence $O_1^n, O_2^n, \ldots$ of open n-cells in M^n with $O_j^n \subset O_{j+1}^n, j = 1, 2, \ldots$, and with

$$M^n = \bigcup_{j=1}^{\infty} O_j^n.$$

If a topological space is the monotone union of open n-cells, then it is an open n-cell [*1*]. We conclude that M^n is homeomorphic to the n-dimensional euclidean space $\mathbb{R}^n$.

Corollary 3.1. *An n-connected, at infinity* 1*-connected, open n-dimensional topological manifold is homeomorphic to the n-dimensional euclidean space* $\mathbb{R}^n$ *for* $n \geqq 5$.

Proof. Lemma 2.1 and Theorem 3.2.

Corollary 3.2. *If* M_1 *and* M_2 *are contractible open topological manifolds of dimensions* n_1 *and* n_2*, where* $n_1 \geqq 1$, $n_2 \geqq 1$, *and* $n_1 + n_2 \geqq 5$, *then* $M_1 \times M_2$ *is homeomorphic to the n-dimensional euclidean space* $\mathbb{R}^n$.

Proof. Lemma 2.2 and Theorem 3.2.

Corollary 3.3. *If* M^n *is a contractible open n-dimensional topological manifold and if* $n \geqq 4$, *then* $M^n \times \mathbb{R}$ *is homeomorphic to the* $(n+1)$*-dimensional euclidean space* $\mathbb{R}^{n+1}$. *If* M^3 *is a contractible open* 3*-dimensional topological manifold, then* $M^3 \times \mathbb{R}^2$ *is homeomorphic to the* 5*-dimensional euclidean space* $\mathbb{R}^5$. (Since each 3-dimensional topological manifold has a piecewise linear structure, this last result is of course contained in the piecewise linear version of this corollary proved in [*9*]).

Remarks. Lemma 3.1, Theorems 3.1 and 3.2, and the Corollaries 3.1, 3.2, and 3.3 in the category of topological manifolds hold also in the

category of piecewise linear manifolds. Our proofs hold in both categories (using the piecewise linear version of the Engulfing Theorem [9] in the piecewise linear case). The piecewise linear version of Theorem 3.2 and of the Corollaries 3.1, 3.2, and 3.3 were proved in [9]. The piecewise linear form of Theorem 3.1, however, does not seem to be explicitly stated in the literature:

Let M^n be a contractible open n-dimensional piecewise linear manifold. If $n \geqq 5$, then M^n can be covered with two open piecewise linear n-cells.

4. Application

Let $P(n)$ denote the statement of the topological Poincaré conjecture in dimension n: If an n-dimensional topological manifold is a homotopy n-sphere, then it is a topological n-sphere. $P(n)$ was proved for $n \neq 3, 4$ in [7].

Let $E(n)$ denote the following statement for $n \geqq 3$: A contractible, at infinity 1-connected, open n-dimensional topological manifold is homeomorphic to the n-dimensional euclidean space $\mathbb{R}^n$. We proved $E(n)$ for $n \geqq 5$.

It follows from Corollary 2 of [10] that $P(3)$ and $E(3)$ are equivalent. Question: Are $E(4)$ and $P(4)$ equivalent?

The conjecture $E(n)$ is actually stronger than the conjecture $P(n)$. If $E(n)$, $n \geqq 3$, is true, then also $P(n)$. (Namely, suppose that M^n is an n-dimensional topological manifold which has the homotopy type of an n-sphere. M^n is then a closed topological manifold. Removing a point p, gives the contractible, at infinity 1-connected, open topological manifold $M^n - \{p\}$. Since $E(n)$ shall be true, $M^n - \{p\}$ is homeomorphic to $\mathbb{R}^n$. The one-point compactification of $M^n - \{p\}$ is homeomorphic to M^n, and hence M^n is homeomorphic to the n-sphere).

Theorem 4.1. *Let M^n, $n \geqq 3$, be an n-dimensional topological manifold, and suppose that M^n is a homotopy n-sphere. If $E(n+1)$ is true, then the suspension $S(M^n)$ of M^n is a topological $(n+1)$-sphere.*

Proof. Let B^n be a flat n-cell in M^n. Then $D^n = M^n - \mathrm{int}(B^n)$ is a contractible compact n-dimensional topological manifold with a topological $(n-1)$-sphere as boundary, and $\mathrm{int}(D^n)$ is a contractible open n-dimensional topological manifold. By Lemma 2.2, $\mathrm{int}(D^n \times I) = \mathrm{int}(D^n) \times \mathrm{int}(I)$, $I = [0, 1]$ the unit interval, is a contractible, at infinity 1-connected, open $(n+1)$-dimensional topological manifold. Since $E(n+1)$ shall be true, $\mathrm{int}(D^n \times I)$ is homeomorphic to $\mathbb{R}^{n+1}$. It was shown in [4] that if two compact topological manifolds have homeomorphic interiors, then the suspensions of their boundaries are homeomorphic. The suspension

$S(\partial(D^n \times I))$ is therefore homeomorphic to the suspension $S(\partial I^{n+1})$. Thus $S(\partial(D^n \times I))$ is a topological $(n+1)$-sphere.

The topological n-sphere $S^n = S(\partial D^n \times 0)$ is locally flat in $S(\partial(D^n \times I))$ up to the north pole and the south pole of the suspension. By [3], S^n decomposes $S(\partial(D^n \times I))$ into two closed $(n+1)$-cells. We conclude that the suspension $S(D^n)$ is a topological $(n+1)$-cell, and that the suspension $S(M^n)$ is a topological $(n+1)$-sphere.

Corollary 4.1. *If the 4-dimensional topological manifold M^4 is a homotopy 4-sphere, then the suspension $S(M^4)$ of M^4 is a topological 5-sphere.*

Proof. Theorems 3.2 and 4.1.

Corollary 4.2. *Let M^3 be a 3-dimensional topological manifold, and suppose that M^3 is a homotopy 3-sphere. If $E(4)$ is true, then the suspension $S(M^3)$ of M^3 is a topological 4-sphere.*

Proof. Theorem 4.1.

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